India's Clean Electricity Trajectory:

Will the 500GW Target Be Met by 2030?

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Introduction

Energy dictates a lot of what we do on Earth. The energy sector is responsible for ~75% of global emissions, of which nearly 30% is electricity (and heat) generation (Ge et al., 2024). It logically follows that countries worldwide are scrambling to reduce these emissions. This is done primarily by increasing non-fossil fuel (hereafter NFF) electricity generation and decreasing fossil fuel usage. Countries have set ambitious goals, aiming to achieve X% total capacity by the year Y, and India is no exception.

India has set the goal of having 500GW NFF energy (electricity) generation capacity by 2030 (Kumar, 2025). India has achieved a few things as of 2026, as per the Ministry of New and Renewable Energy (Sreejith, 2025). Firstly, 50% of cumulative electricity generation capacity comes from NFF sources; this target was achieved 5 years ahead of schedule in June 2025. Secondly, of the 500GW goal, 262.7GW installed capacity has been reached, which is positive news. This paper aims to fully clarify some confusing parts in this news and understand its implications.

Clarifications

To understand what is happening, a few ideas require examination. Firstly, what is the difference between energy and electricity? Energy is a broad term used to describe electricity generation, thermal energy (for cooking or heating), transportation fuels, industrial processes, etc. Electricity is simply one form of energy, describing the movement of electrons. India's 2030 500GW goal is for NFF electricity, which includes solar, hydro (large and small), wind, nuclear, and bio-power.

Now for a few technical terms:

Installed capacity (Watts (W)): this is the maximum power a plant could produce if it ran 24/7 under ideal conditions. For example, a solar farm rated 500MW has the capacity to generate 500MW of electricity if the Sun were around the entire day, at noon position, without clouds. Installed capacity does not describe how much energy is actually produced, which leads us to—

Capacity factor (%): this is the ratio of actual produced electricity with respect to installed capacity over a certain period of time (usually a year = 8760 hours). Our solar farm may have a

capacity factor of 20% because the sun does not shine at night, it gets cloudy sometimes, etc.; i.e. the 500MW capacity power plant actually produces about 20% of 500MW, which is the—

Electricity generation (Watt-hours (Wh)): this is the actual electricity generated over a certain period of time. Put simply: Electricity Generated = Installed Capacity × Capacity Factor × Time Period. Therefore, our solar farm produced $500 \times 0.2 \times 8760 = 876$GWh. Comparatively, a 100MW coal power plant running at 100% capacity would generate the same amount.

India's renewable energy goals describe electricity, not cumulative energy. Renewable energy implementation requires outfitting multiple sectors, and is difficult to measure. Installing and measuring renewable electricity is far tractable a goal. While India has an installed renewable capacity of 262GW (~50%), actual renewable electricity generated is far lesser, at about 25% (Koshy, 2025). The 50% achievement is certainly commendable, but it is important to understand the distinction between capacity and generation, as generation usually lags behind capacity. Hence we reach the questions, is India on track for its 500GW renewable electricity capacity by 2030 goal? When will renewable electricity generation overtake fossil fuels?

Process

This paper aims to predict the answers for the above two questions by utilising time-series linear and exponential regression (refer Appendix A for formulae). The regressions will be limited to electricity capacity and generation within India, and will not venture towards energy. The linear model assumes that the past trend of capacity addition continues, whereas the exponential model incorporates a scenario wherein electricity capacity and generation growth increases annually. This exponential model implies continued acceleration and is a simplified upper-bound scenario. This model does not feature a sensitivity analysis, choosing instead to simply answer the previous questions. Factors such as possible policy changes, climate effects, and technological innovation have not been taken into account, being out of scope of this technical paper.

Data for NFF capacity is acquired from Bharat Energy (NITI Aayog, 2026), while the NFF generation data is acquired from Our World In Data (Ritchie & Rosado, 2025); refer Appendix A for data tables. The regression models applied will utilise numbers from 2014 to 2025. The regression models incorporate NFF/low-carbon source electricity, which includes solar,

cumulative hydro, wind, nuclear, and bio-power.

Results

With these constraints, a modelling code was written in Python (refer Appendix B):

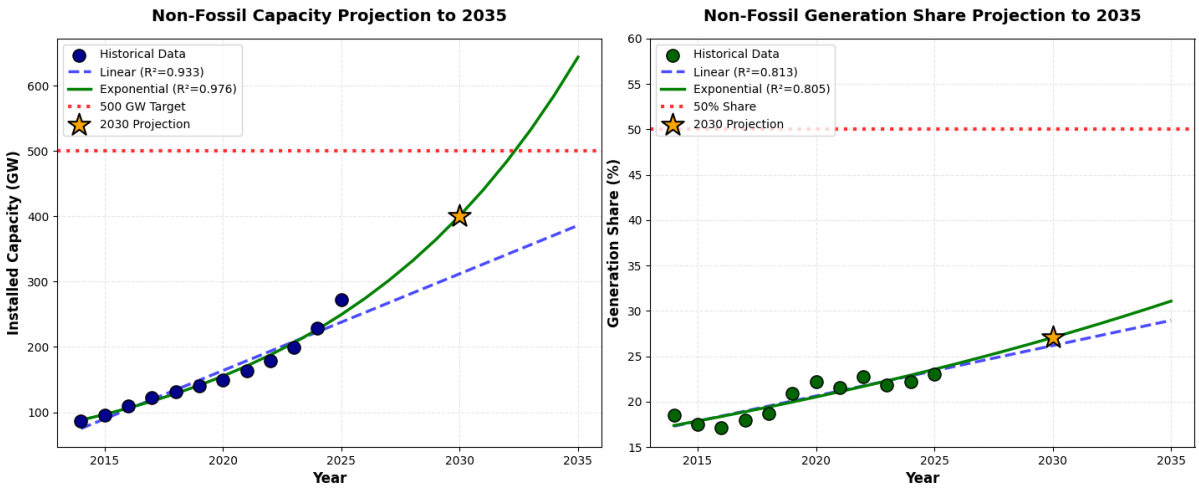

Figure 1: Non-Fossil Capacity and Generation Share Projections to 2035

As per the models, India is not on track to meet the 2030 500GW NFF electricity capacity goal. By 2030, the linear and exponential models predict that India will have reached 311.9GW and 400.6GW capacity respectively. Given the accelerated pace displayed by India in 2025, having added 44.5GW capacity by November (Sreejith, 2025), we can plausibly consider the exponential model to be more in-line with expectations. However that model falls short of the goal by roughly 100GW, with the goal actually being achieved by 2033 (a slight delay). As NFF electricity generation share remains below 30% in both models, generation numbers show cause for concern as NFF generation will not be overtaking fossil-fuel sources until around 2050.

Conclusion

Capacity for NFF electricity is an important metric, as a country needs the capacity before it can get to generation. This paper's analysis shows that India is slightly behind the 500GW goal in this department, but this delay is manageable. Further acceleration in NFF power plants over the next four years can easily tide India over its 2030 goal.

The greater point of focus appears to be NFF generation share, that sits below 30% now and remains so half a decade later. Capacity is important, but until that translates into generation,

the environmental cost of emissions remains dangerous (fossil-fuel generation dominates). A possible cause for this shortfall is intermittency, a physical constraint wherein NFF sources cannot generate electricity at full capacity 24/7. India's NFF capacity is mostly intermittent solar (NITI Aayog, 2026), and hence its generation is greatly vulnerable to weather, climate, and the night. Another possible cause is dispatchability, the ability to produce electricity on demand. This can be augmented by utilisation of dispatchable NFF sources which are more protected against intermittency, such as nuclear or hydropower. Acceleration of construction of such NFF sources can greatly augment electricity generation share, not just capacity.

Meeting the 2030 capacity goal through policy changes is quite achievable, especially through the relevant policy changes. Having NFF generation overtake fossil-fuel generation, requires not just policy, but public demand, awareness, and education.

References

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Appendix A: Model and Data

Disclaimer: AI tools were used to assist in formulating the mathematical structure of the carbon payback model and generating the initial Python code for calculations. All assumptions, parameter choices, interpretation of results, and written analysis are of the author's. AI was used as a computational aid, not for generating any essay content.

The general linear regression formula:

$$y = \beta_0 + \beta_1 x$$

The exponential linear regression formula:

$$y = ae^{bx}$$

Data:

Year	Non-Fossil Capacity (GW)	Generation Share (%)
2014	87.0	18.5
2015	95.7	17.5
2016	109.8	17.1
2017	122.7	18.0
2018	131.6	18.7
2019	140.7	20.9
2020	148.8	22.2
2021	163.4	21.6
2022	178.8	22.8
2023	198.8	21.8
2024	228.3	22.2
2025	272.0	23.0

2030 Results:

Non-Fossil Capacity:

- Linear Model: 311.9 GW
- Exponential Model: 400.6 GW
- Target: 500.0 GW
- Gap (Linear): 188.1 GW
- Gap (Exponential): 99.4 GW

Non-Fossil Generation Share:

- Linear Model: 26.2%
- Exponential Model: 27.1%

Year 500 GW reached (Exponential): 2033

Appendix B: Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

# Data (2014-2025)
years_raw = np.array([...])
base_year = 2014
years = (years_raw - base_year).reshape(-1,1)

nonfossil_capacity_mw = np.array([...])
nonfossil_capacity = nonfossil_capacity_mw / 1000  # Convert to GW

nonfossil_generation = np.array([...])

# Linear regression function
def linear_regression(x, y):
    model = LinearRegression()
    model.fit(x, y)
    y_pred = model.predict(x)
    r2 = r2_score(y, y_pred)
    return model, y_pred, r2

# Exponential regression function
def exponential_regression(x, y):
    log_y = np.log(y)
    model = LinearRegression()
    model.fit(x, log_y)
    log_pred = model.predict(x)
```

```
    y_pred = np.exp(log_pred)
    r2 = r2_score(y, y_pred)
    return model, y_pred, r2

# Fit models
cap_lin_model, cap_lin_pred, cap_lin_r2 = \
    linear_regression(years, nonfossil_capacity)
cap_exp_model, cap_exp_pred, cap_exp_r2 = \
    exponential_regression(years, nonfossil_capacity)

# Project to 2030 and beyond
future_years_raw = np.arange(2014, 2036)
future_years = (future_years_raw - base_year).reshape(-1,1)

cap_exp_future = np.exp(cap_exp_model.predict(future_years))

# Calculate 2030 values
year_2030 = np.array([[2030 - base_year]])
cap_2030_exp = np.exp(cap_exp_model.predict(year_2030))[0]

print(f"2030 Capacity (Exponential): {cap_2030_exp:.1f} GW")
print(f"Gap from 500 GW: {500 - cap_2030_exp:.1f} GW")

# [Plotting code omitted for brevity]
```